

1 Plant and Control Law

1.1 Continuous-time plant

$$\gamma(x) \dot{x} = f_{dx} + f_{Dx} + f_T \quad (1)$$

1.2 Discrete-time plant (zero-order hold)

$$a_x[k] \equiv \Delta t / \gamma(x[k]) \quad (2)$$

$$x_D[k] \equiv a_x[k] f_{Dx}[k] \quad (3)$$

$$x[k+1] = x[k] + a_x[k] (f_{dx}[k] + f_T[k]) + x_D[k] \quad (4)$$

1.3 Thermal force statistics

$f_T[k]$ is zero-mean Gaussian white with variance

$$\sigma_{f_T}^2[k] = \frac{4k_B T \gamma(x[k])}{\Delta t} \implies \text{Var}(a_x[k] f_T[k]) = 4k_B T a_x[k]. \quad (5)$$

1.4 Control law

$$f_{dx}[k] = \hat{a}_x^{-1}[k] \left[(x_d[k+1] - x_d[k]) + (1 - \lambda_c) \delta \hat{x}[k] - \hat{x}_D[k] \right] \quad (6)$$

In the augmented state of §3, $\delta \hat{x}[k] = \delta \hat{x}_3[k]$ (due to the two-step measurement delay; see §3).

2 IIR Variance Estimator for $a_x[k]$

This section details the infinite impulse response (IIR) variance estimator that produces the auxiliary measurement a_{xm} from the measured tracking error. The pipeline combines low-pass (LP) and high-pass (HP) stages.

2.1 Tracking error and high-pass filter

$$\delta x_m[k] = x_d[k-2] - x_m[k] \quad (7)$$

$$\delta x_{md}[k] = (1 - a_{pd}) \delta x_{md}[k-1] + a_{pd} \delta x_m[k] \quad (8)$$

$$\delta x_{mr}[k] = \delta x_m[k] - \delta x_{md}[k] \quad (9)$$

Only δx_{mr} is driven by $a_x f_T$; δx_{md} holds the trajectory-tracking mean.

2.2 Single-sample variance estimator

$\sigma^2(X) = \text{E}[X^2] - (\text{E}[X])^2$ with two IIR sub-estimators:

$$\delta x_{mrd}[k] = (1 - a_{prd}) \delta x_{mrd}[k-1] + a_{prd} \delta x_{mr}[k] \quad (10)$$

$$\overline{(\delta x_{mr})^2}[k] = (1 - a_{cov}) \overline{(\delta x_{mr})^2}[k-1] + a_{cov} (\delta x_{mr}[k])^2 \quad (11)$$

$$\sigma_{\delta x_{mr}}^2[k] = \overline{(\delta x_{mr})^2}[k] - (\delta x_{mrd}[k])^2 \quad (12)$$

2.3 Role of the three IIR coefficients

coef	role	analyzed in
a_{pd}	LP pole of δx_m ; defines δx_{mr}	§5 (C_{dpmr}, C_n)
a_{prd}	LP pole of δx_{mr}	§6 (β)
a_{cov}	LP pole of $(\delta x_{mr})^2$	bias-free; estimator variance only

2.4 Link between $\sigma_{\delta x_{mr}}^2$ and a_x

$$\boxed{\sigma_{\delta x_{mr}}^2[k] \approx \beta (C_{\text{dpmr}} \cdot 4k_B T a_x[k] + C_n \sigma_n^2)} \quad (13)$$

(C_{dpmr}, C_n) derived in §5; β in §6.

$$a_{xm}[k] = \frac{\sigma_{\delta x_{mr}}^2[k]/\beta - C_n \sigma_n^2}{C_{\text{dpmr}} \cdot 4k_B T} \quad (14)$$

3 Kalman Filter (KF)

3.1 State vector and process model

$$\mathbf{x}[k] = [\delta x_1, \delta x_2, \delta x_3, x_D, \delta x_D, a_x, \delta a_x]^T[k] \quad (15)$$

$$\mathbf{x}[k+1] = \mathbf{F}_e \mathbf{x}[k] + \mathbf{q}[k] \quad (16)$$

$$\mathbf{F}_e = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -f_{dx}[k] & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2 Measurement model

$$\mathbf{y}[k] = \mathbf{H} \mathbf{x}[k] + \mathbf{v}[k] = \begin{bmatrix} \delta x_m[k] \\ a_{xm}[k] \end{bmatrix} \quad (17)$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

3.3 Kalman filter recursions

$$\mathbf{L}[k] = \mathbf{P}_f[k] \mathbf{H}^T (\mathbf{H} \mathbf{P}_f[k] \mathbf{H}^T + \mathbf{R})^{-1} \quad (18)$$

$$\mathbf{P}[k] = (\mathbf{I} - \mathbf{L}[k] \mathbf{H}) \mathbf{P}_f[k] \quad (19)$$

$$\mathbf{P}_f[k+1] = \mathbf{F}_e \mathbf{P}[k] \mathbf{F}_e^T + \mathbf{Q} \quad (20)$$

$$\hat{\mathbf{x}}[k+1 | k] = \mathbf{F}_e \hat{\mathbf{x}}[k | k-1] + \mathbf{F}_e \mathbf{L}_{ss} (\mathbf{y}[k] - \mathbf{H} \hat{\mathbf{x}}[k | k-1]) \quad (21)$$

5 Closed-Loop Steady-State Variance: C_{dpmr} and C_n

Scope note. The derivation drops the time-varying entry $\mathbf{F}_e(3, 6) = -f_{dx}[k]$ from the $\mathbf{A}$ matrix to obtain a time-invariant system so that Lyapunov analysis applies. Under zero-mean f_{dx} operating conditions (static regulation with $\Delta p_d = 0$, or symmetric oscillatory reference), the dropped $f_{dx} \cdot e_a$ term has $\mathbb{E}[f_{dx} \cdot e_a] \approx 0$ and contributes variance of order $\sigma_f^2 \sigma_e^2$, which is negligible relative to the dominant thermal term $4k_B T a_x$ (i.e. $\sigma_f^2 \sigma_e^2 / (4k_B T a_x) \ll 1$). Empirical variance matches the derived C_{dpmr} at all operating points tested (free-space, near-wall static, dynamic sweep).

5.1 Augmented state and closed-loop dynamics

$$\mathbf{e}[k] = [\delta x_1, \delta x_2, \delta x_3, e_{x1}, e_{x2}, e_{x3}, e_{xD}, e_{\delta xD}, e_a, e_{\delta a}, \delta x_{md}[k-1]]^T[k] \in \mathbb{R}^{11}$$

$$\mathbf{e}[k+1] = \mathbf{A} \mathbf{e}[k] + \mathbf{b}_T(a_x[k] f_T[k]) + \mathbf{b}_n n[k] \quad (22)$$

$$\mathbf{A} = \left[\begin{array}{ccc|ccccccc|c} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & 0 & 0 & 1-\lambda_c & -1 & 0 & 0 & 0 \\ \hline \mathbf{0}_{7 \times 3} & & & \mathbf{A}_e & & & & & \mathbf{0}_{7 \times 1} & \\ \hline a_{pd} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-a_{pd} \end{array} \right]$$

where $\mathbf{A}_e \equiv \mathbf{F}_e(\mathbf{I} - \mathbf{L}_{ss}\mathbf{H})$ is the steady-state KF estimation-error dynamics (§3.3).

$$\mathbf{b}_T = [0, 0, -1, 0, 0, -1, 0, 0, 0, 0, 0]^T, \quad \mathbf{b}_n = [0, 0, 0, -\mathbf{F}_e \mathbf{L}_{ss}(:, 1), -a_{pd}]^T$$

5.2 Lyapunov decomposition

Define $\Sigma[k] \equiv \mathbb{E}[\mathbf{e}[k]\mathbf{e}[k]^T]$. Since $f_T[k]$ and $n[k]$ are independent zero-mean white processes, Σ decomposes linearly:

$$\Sigma[k] = 4k_B T a_x[k] \Sigma_T + \sigma_n^2 \Sigma_n \quad (23)$$

$$\Sigma_T = \mathbf{A} \Sigma_T \mathbf{A}^T + \mathbf{b}_T \mathbf{b}_T^T \quad (24)$$

$$\Sigma_n = \mathbf{A} \Sigma_n \mathbf{A}^T + \mathbf{b}_n \mathbf{b}_n^T \quad (25)$$

5.3 Variance of δx_{mr}

Substituting the §2.1 LP recursion into the HP residual definition, with $\delta x_m[k] = \delta x_1[k] - n[k]$:

$$\delta x_{mr}[k] = (1 - a_{pd})(\delta x_1[k] - \delta x_{md}[k-1] - n[k]) \quad (26)$$

Variance ($n[k]$ independent of $\mathbf{e}[k]$):

$$\sigma_{\delta x_{mr}}^2[k] = (1 - a_{pd})^2 \left[(\Sigma[k])_{1,1} - 2(\Sigma[k])_{1,11} + (\Sigma[k])_{11,11} + \sigma_n^2 \right] \quad (27)$$

5.4 C_{dpmr} and C_n

Substituting $\Sigma[k] = 4k_B T a_x[k] \Sigma_T + \sigma_n^2 \Sigma_n$ into the §5.3 variance formula and matching against the target $\sigma_{\delta x_{mr}}^2 = C_{\text{dpmr}} \cdot 4k_B T a_x + C_n \sigma_n^2$:

Σ_T and Σ_n are the discrete Lyapunov solutions — equivalently, geometric series in $\mathbf{A}$ — driven by $\mathbf{b}_T$ and $\mathbf{b}_n$ respectively:

$$\begin{aligned}\Sigma_T(\lambda_c, a_{pd}; \mathbf{Q}, \mathbf{R}) &= \sum_{k=0}^{\infty} \mathbf{A}^k \mathbf{b}_T \mathbf{b}_T^T (\mathbf{A}^T)^k \\ \Sigma_n(\lambda_c, a_{pd}; \mathbf{Q}, \mathbf{R}) &= \sum_{k=0}^{\infty} \mathbf{A}^k \mathbf{b}_n \mathbf{b}_n^T (\mathbf{A}^T)^k\end{aligned}$$

(solutions of $\mathbf{A}\Sigma_T\mathbf{A}^T + \mathbf{b}_T\mathbf{b}_T^T = \Sigma_T$ and $\mathbf{A}\Sigma_n\mathbf{A}^T + \mathbf{b}_n\mathbf{b}_n^T = \Sigma_n$; $\mathbf{A}, \mathbf{b}_T, \mathbf{b}_n$ constructed per §5.1.)

Therefore:

$$\boxed{C_{\text{dpmr}}(\lambda_c, a_{pd}; \mathbf{Q}, \mathbf{R}) = (1 - a_{pd})^2 [(\Sigma_T)_{1,1} - 2(\Sigma_T)_{1,11} + (\Sigma_T)_{11,11}]}$$
 (28)

$$\boxed{C_n(\lambda_c, a_{pd}; \mathbf{Q}, \mathbf{R}) = (1 - a_{pd})^2 [(\Sigma_n)_{1,1} - 2(\Sigma_n)_{1,11} + (\Sigma_n)_{11,11} + 1]}$$
 (29)

The +1 in C_n captures the direct contribution of the current $n[k]$ via the explicit $-n$ term in §5.3; the (Σ_n) selector captures past $n[k]$'s propagated through the state dynamics $\mathbf{e}[k]$. With $(\mathbf{Q}, \mathbf{R})$ fixed at design time, the LUT is built offline on a (λ_c, a_{pd}) grid:

```
for each  $(\lambda_c, a_{pd})$  in design grid:
  build  $\mathbf{A}$  and  $\mathbf{b}_n$  per §5.1 from  $(\lambda_c, a_{pd}; \mathbf{Q}, \mathbf{R}, \mathbf{H})$ 
   $\Sigma_T \leftarrow \text{dlyap}(\mathbf{A}, \mathbf{b}_T \mathbf{b}_T^T)$ 
   $\Sigma_n \leftarrow \text{dlyap}(\mathbf{A}, \mathbf{b}_n \mathbf{b}_n^T)$ 
   $C_{\text{dpmr}}[\lambda_c, a_{pd}] \leftarrow (1 - a_{pd})^2 [(\Sigma_T)_{1,1} - 2(\Sigma_T)_{1,11} + (\Sigma_T)_{11,11}]$ 
   $C_n[\lambda_c, a_{pd}] \leftarrow (1 - a_{pd})^2 [(\Sigma_n)_{1,1} - 2(\Sigma_n)_{1,11} + (\Sigma_n)_{11,11} + 1]$ 
```

6 IIR Finite-Sample Bias Factor β

6.1 Bias source

The Section 2 IIR algorithm computes $\sigma_{\delta x_{mr}}^2[k] = \overline{(\delta x_{mr})^2}[k] - (\delta x_{mrd}[k])^2$. In stationary steady state (δx_{mr} zero-mean):

$$\mathbb{E}[\overline{(\delta x_{mr})^2}] = \text{Var}(\delta x_{mr})$$
 (30)

$$\mathbb{E}[(\delta x_{mrd})^2] = \text{Var}(\delta x_{mrd})$$
 (31)

so the IIR output systematically underestimates $\text{Var}(\delta x_{mr})$ by the amount $\text{Var}(\delta x_{mrd}) > 0$. Define the bias factor β as the ratio of the IIR estimator's expected output $\mathbb{E}[\sigma_{\delta x_{mr}}^2]$ to the true variance $\text{Var}(\delta x_{mr})$:

$$\beta \equiv \frac{\mathbb{E}[\sigma_{\delta x_{mr}}^2]}{\text{Var}(\delta x_{mr})} = 1 - \frac{\text{Var}(\delta x_{mrd})}{\text{Var}(\delta x_{mr})}$$
 (32)

6.2 $\text{Var}(\delta x_{mr})$ and autocorrelation $\rho(L)$

To evaluate β , we need the denominator $\text{Var}(\delta x_{mr})$ and the temporal autocorrelation $\rho(L)$ used for the numerator.

For the noise-free case ($\sigma_n^2 = 0$), §5's selector form gives

$$\text{Var}(\delta x_{mr}) = (1 - a_{pd})^2 [(\Sigma_T)_{1,1} - 2(\Sigma_T)_{1,11} + (\Sigma_T)_{11,11}] \cdot 4k_B T a_x.$$
 (33)

The lag- L autocorrelation is defined as

$$\rho(L) \equiv \text{Cov}(\delta x_{mr}[k], \delta x_{mr}[k-L]) / \text{Var}(\delta x_{mr}). \quad (34)$$

Since $\mathbf{A}^L \mathbf{\Sigma}_T$ is the lag- L cross-covariance of $\mathbf{e}[k]$, applying the same selector (the common $(1 - a_{pd})^2$ and $4k_B T a_x$ prefactors cancel in the ratio) yields

$$\rho(L) = \frac{(\mathbf{A}^L \mathbf{\Sigma}_T)_{1,1} - 2(\mathbf{A}^L \mathbf{\Sigma}_T)_{1,11} + (\mathbf{A}^L \mathbf{\Sigma}_T)_{11,11}}{(\mathbf{\Sigma}_T)_{1,1} - 2(\mathbf{\Sigma}_T)_{1,11} + (\mathbf{\Sigma}_T)_{11,11}}. \quad (35)$$

6.3 $\text{Var}(\delta x_{mrd})$ closed form

Unrolling the §2.2 LP recursion converts $\delta x_{mrd}[k]$ into an explicit weighted sum of past δx_{mr} , so that $\text{Var}(\delta x_{mrd})$ can be expressed via $\rho(L)$:

$$\delta x_{mrd}[k] = a_{prd} \sum_{j=0}^{\infty} (1 - a_{prd})^j \delta x_{mr}[k-j] \quad (36)$$

Take variance, split diagonal $j=l$ from off-diagonal $j \neq l$, and use $\sum_{j=0}^{\infty} (1 - a_{prd})^{2j} = 1 / [a_{prd}(2 - a_{prd})]$:

$$\text{Var}(\delta x_{mrd}) = \frac{a_{prd}}{2 - a_{prd}} \text{Var}(\delta x_{mr}) \left[1 + 2 \sum_{m=1}^{\infty} \rho(m) (1 - a_{prd})^m \right] \quad (37)$$

6.4 Bias factor β

Substituting the $\text{Var}(\delta x_{mrd})$ expression into the β definition:

$$\boxed{\beta(\lambda_c, a_{pd}, a_{prd}; \mathbf{Q}, \mathbf{R}) = 1 - \frac{a_{prd}}{2 - a_{prd}} \left[1 + 2 \sum_{m=1}^{\infty} \rho(m) (1 - a_{prd})^m \right]} \quad (38)$$

The bracket is the *autocorrelation amplification*: = 1 for white δx_{mr} , > 1 for positively-correlated.

7 Q/R Design

This section derives each non-zero entry of the Kalman filter (KF) process-noise $\mathbf{Q}$ and measurement-noise $\mathbf{R}$ from physical first principles.

7.1 Zero elements of $\mathbf{Q}$

$Q(i, i)$ represents the process noise entering state i . Under *positioning* (probe held at fixed h_{init}), most states have no driver:

- $\delta x_1, \delta x_2$: past values of δx_3 , shifted by delay.
- $x_D, \delta x_D$: disturbance is modeled as zero ($x_D \equiv 0$).
- δa_x : $a = a_{\text{nom}}/c_{\text{axis}}(\bar{h})$ is constant at fixed h , so $\delta a \equiv 0$.

Only two entries are non-zero: $Q(3, 3)$ (thermal force injection into δx_3) and $Q(6, 6)$ (a small positive regularization value, see below).

7.2 Non-zero elements of Q

$Q(3,3)$ — thermal force injection.

Per-axis thermal-force variance from the Einstein relation with wall-corrected drag:

$$\text{Var}(f_{T,\text{axis}}) = \frac{4k_B T \gamma_N c_{\text{axis}}(\bar{h})}{\Delta t}. \quad (39)$$

Multiplying by a_{axis}^2 gives the thermal injection into δx_3 :

$$\text{Var}(a_{\text{axis}} f_T) = 4k_B T \cdot a_{\text{axis}} = a_{\text{ratio}} \cdot \sigma_{\text{dXT}}^2, \quad (40)$$

with $\sigma_{\text{dXT}}^2 \equiv 4k_B T a_{\text{nom}}$, $a_{\text{nom}} \equiv \Delta t / \gamma_N$, $a_{\text{ratio}} \equiv a_{\text{axis}} / a_{\text{nom}}$. Hence

$$\boxed{Q(3,3) = a_{\text{ratio}} \cdot \sigma_{\text{dXT}}^2} \quad (41)$$

$Q(6,6)$ — a small positive value ϵ .

Under positioning, a is constant, so physically $Q(6,6) = 0$. But setting $Q(6,6) = 0$ together with $Q(7,7) = 0$ makes the filter assume a never varies: the discrete algebraic Riccati equation (DARE) then solves to $\mathbf{L}_{ss}(6,:) = 0$, meaning the filter's gain on the a_{xm} measurement is zero — a_{xm} gets ignored entirely. To prevent this, add a small positive floor:

$$Q(6,6) = \epsilon \cdot \sigma_{\text{dXT}}^2, \quad \epsilon \ll 1. \quad (42)$$

This tells the filter to allow a tiny slow drift in a , restoring a small non-zero $\mathbf{L}_{ss}(6,:)$ so that a_{xm} can slowly update $\hat{a}$. ϵ is chosen small enough to preserve the scalar Riccati approximation used in §8.3.

7.3 Measurement noise R (per axis)

$R(1,1)$ — **sensor noise on δx_m** . The position measurement has per-axis sensor-noise variance $\sigma_{n,\text{axis}}^2$. Normalizing by the thermal unit:

$$\boxed{R(1,1)_{\text{axis}} = \sigma_{n,\text{axis}}^2 / \sigma_{\text{dXT}}^2} \quad (43)$$

Per-axis is essential because different axes' tracking algorithms have σ_n values spanning orders of magnitude.

$R(2,2)$ — sampling noise on a_{xm} .

a_{xm} is not a direct measurement — it is the output of the IIR variance estimator (§2.4). Its finite-sample noise variance takes the form

$$\text{Var}(n_a) = \chi_{\text{sq}} \cdot \rho_a \cdot a_{\text{axis}}^2, \quad (44)$$

built from three factors:

- $\chi_{\text{sq}} \equiv 2a_{\text{cov}} / (2 - a_{\text{cov}})$: exponential moving average (EMA) finite-sample factor.
- $\rho_a \equiv 1 + 2 \sum_{L \geq 1} \rho^2(L) (1 - a_{\text{cov}})^L$: autocorrelation amplification factor (≥ 1).
- a_{axis}^2 : signal scale (a_{xm} scales with a).

The squared ρ^2 (vs. §6.4's linear ρ) arises because the EMA acts on $(\delta x_{mr})^2$: Wick/Isserlis gives the lag- L autocorrelation of the squared Gaussian process as $\rho^2(L)$.

The $\rho(L)$ used here has the same selector structure as §6.2 but with $V_{\text{state}} + \sigma_n^2$ in the denominator (§6.2's form is the $\sigma_n^2 = 0$ limit).

Normalizing by the thermal unit:

$$\boxed{R(2,2)_{\text{axis}} = \chi_{\text{sq}} \cdot \rho_{a,\text{axis}} \cdot a_{\text{axis}}^2 / \sigma_{\text{dXT}}^2} \quad (45)$$

7.4 Summary: Q and R as functions of inputs

$$\boxed{Q(3,3)(\bar{h}) = a_{\text{ratio}}(\bar{h}) \cdot \sigma_{\text{dXT}}^2} \quad (46)$$

$$\boxed{Q(6,6)(\epsilon) = \epsilon \cdot \sigma_{\text{dXT}}^2} \quad (47)$$

$$\boxed{R(1,1)_{\text{axis}}(\sigma_{n,\text{axis}}^2) = \sigma_{n,\text{axis}}^2 / \sigma_{\text{dXT}}^2} \quad (48)$$

$$\boxed{R(2,2)_{\text{axis}}(\bar{h}, a_{\text{cov}}, \rho_{a,\text{axis}}) = \chi_{\text{sq}} \cdot \rho_{a,\text{axis}} \cdot a_{\text{axis}}^2(\bar{h}) / \sigma_{\text{dXT}}^2} \quad (49)$$

Note on $\rho_{a,\text{axis}}$ dependency: via $\mathbf{A}_{\text{aug}}$ and Σ_{aug} , $\rho_{a,\text{axis}}$ itself is a function of $(\lambda_c, a_{pd}, a_{\text{cov}}, \bar{h}, \epsilon, \sigma_{n,\text{axis}}^2)$ via a self-consistent fixed point. Hence $R(2,2)_{\text{axis}}$ ultimately depends on the same input set.

8 Theoretical Performance

Combining the Q/R design of §7 with the augmented Lyapunov framework of §5 gives closed-form predictions for the tracking-error variance and the motion-gain estimation variance per axis.

8.1 Three-channel augmented covariance

The two-channel decomposition of §5.2 is extended with a third channel for the a_{xm} sampling noise. Since f_T , n_p , and n_a are mutually independent white noises, the augmented covariance splits linearly:

$$\boxed{\Sigma_{\text{aug}} = a_{\text{ratio}} \sigma_{\text{dXT}}^2 \Sigma_T + \sigma_n^2 \Sigma_n + \chi_{\text{sq}} \rho_a a^2 \Sigma_{na}} \quad (50)$$

Each Σ_X solves the unit-variance Lyapunov equation $\Sigma_X = \mathbf{A} \Sigma_X \mathbf{A}^T + \mathbf{b}_X \mathbf{b}_X^T$ with input vector

$$\mathbf{b}_T = [0, 0, -1; 0, 0, -1, 0, 0, 0, 0; 0]^T \quad (\text{thermal; §5.1})$$

$$\mathbf{b}_n = [0, 0, 0; -\mathbf{F}_e \mathbf{L}_{ss}(:, 1); -a_{pd}]^T \quad (\text{sensor; §5.1})$$

$$\mathbf{b}_{na} = [0, 0, 0; -\mathbf{F}_e \mathbf{L}_{ss}(:, 2); 0]^T \quad (\text{new: } a_{xm} \text{ channel})$$

$\mathbf{b}_{na}$ mirrors $\mathbf{b}_n$ but uses the second column of $\mathbf{L}_{ss}$ (corresponding to the a_{xm} measurement) and has a zero IIR entry because a_{xm} does not feed the δx_{md} filter.

8.2 Tracking-error variance (per axis)

The first diagonal element of Σ_{aug} (corresponding to δx_1 , the tracking error) gives

$$\boxed{\text{Var}(\delta x_{\text{axis}}) = (\Sigma_{\text{aug}})_{1,1}} \quad (51)$$

with three additive contributions:

- thermal: $a_{\text{ratio}} \sigma_{\text{dXT}}^2 (\Sigma_T)_{1,1}$
- sensor: $\sigma_n^2 (\Sigma_n)_{1,1}$
- a_{xm} sampling: $\chi_{\text{sq}} \rho_a a^2 (\Sigma_{na})_{1,1}$

Per-axis scaling enters through a_{ratio} , $\sigma_{n,\text{axis}}^2$, a_{axis} , and $\rho_{a,\text{axis}}$.

8.3 Motion-gain estimation (per axis)

We evaluate the *coefficient of variation* (CV) of the motion-gain estimate per axis:

$$\text{CV}(\hat{a}_{\text{axis}}) \equiv \text{std}(\hat{a}_{\text{axis}} - a_{\text{axis}}) / a_{\text{axis}}. \quad (52)$$

CV^2 is additive across independent noise sources. Two sources contribute here:

- **KF estimation error:** the KF cannot perfectly track a from noisy a_m .
- **Wall-effect on a_{true} :** $a_{\text{true}}(\bar{h})$ itself varies as $\bar{h}$ fluctuates with the probe's z -error δp_z ; independent of KF.

(Here c_{axis} denotes $c_{\parallel}$ for x, y axes or $c_{\perp}$ for the z axis, as defined in §7.2.) Thus

$$\text{CV}^2(\hat{a}_{\text{axis}}) = \text{CV}_{\text{KF}}^2 + \text{CV}_{\text{wall}}^2. \quad (53)$$

Block A: KF estimation term.

Step 1 — Decoupling. $Q(7, 7) = 0$ (§7.1) decouples the $(a, \delta a)$ KF block; the a -subsystem reduces to a scalar KF tracking a from noisy a_m .

Step 2 — Scalar Riccati gain. The scalar discrete algebraic Riccati equation yields

$$L_{\text{eff}, \text{axis}} = \sqrt{Q(6, 6) / R(2, 2)_{\text{axis}}}. \quad (54)$$

(L_{eff} matches the full DARE gain $\mathbf{L}_{ss}(6, 2)$ when $Q(7, 7) = 0$; the approximation breaks at $Q(7, 7) > 0$.)

Step 3 — Input noise characterization. The KF sees $a_m = a + n_a$ with $\text{Var}(n_a) = \chi_{\text{sq}} \rho_a a_{\text{axis}}^2$ (§7.3). Importantly, n_a is a first-order autoregressive process (AR(1)) with bandwidth a_{cov} , inherited from the IIR EMA.

Step 4 — Cascaded LP. The KF (bandwidth L_{eff}) low-passes the AR(1) noise n_a (bandwidth a_{cov}). A two-state variance analysis (equivalent to integrating the cascaded AR(1) transfer function in the frequency domain) gives, in the small- L_{eff} and small- a_{cov} limit:

$$\text{Var}(\hat{a} - a) / \text{Var}(n_a) = L_{\text{eff}} / (L_{\text{eff}} + a_{\text{cov}}). \quad (55)$$

This replaces the standard $L_{\text{eff}}/2$ result that would apply if n_a were white noise.

Combining Step 3 and Step 4:

$$\boxed{\text{CV}_{\text{KF}, \text{axis}}^2 = \chi_{\text{sq}} \cdot \rho_{a, \text{axis}} \cdot \frac{L_{\text{eff}, \text{axis}}}{L_{\text{eff}, \text{axis}} + a_{\text{cov}}}} \quad (56)$$

Block B: Wall-effect on a_{true} .

$a_{\text{true}}(\bar{h}) = a_{\text{nom}} / c_{\text{axis}}(\bar{h})$ depends on wall distance. Since $\delta \bar{h} = \delta p_z / R$ (up to sign), linearizing gives:

$$\boxed{\text{CV}_{\text{wall}, \text{axis}}^2 = \left(\frac{|dc_{\text{axis}}/d\bar{h}|}{c_{\text{axis}}} \right)^2 \cdot \frac{\text{Var}(\delta p_z)}{R^2}} \quad (57)$$

Total (assuming independence):

$$\boxed{\text{CV}^2(\hat{a}_{\text{axis}}) = \chi_{\text{sq}} \rho_{a, \text{axis}} \frac{L_{\text{eff}}}{L_{\text{eff}} + a_{\text{cov}}} + \left(\frac{|dc_{\text{axis}}/d\bar{h}|}{c_{\text{axis}}} \right)^2 \frac{\text{Var}(\delta p_z)}{R^2}} \quad (58)$$

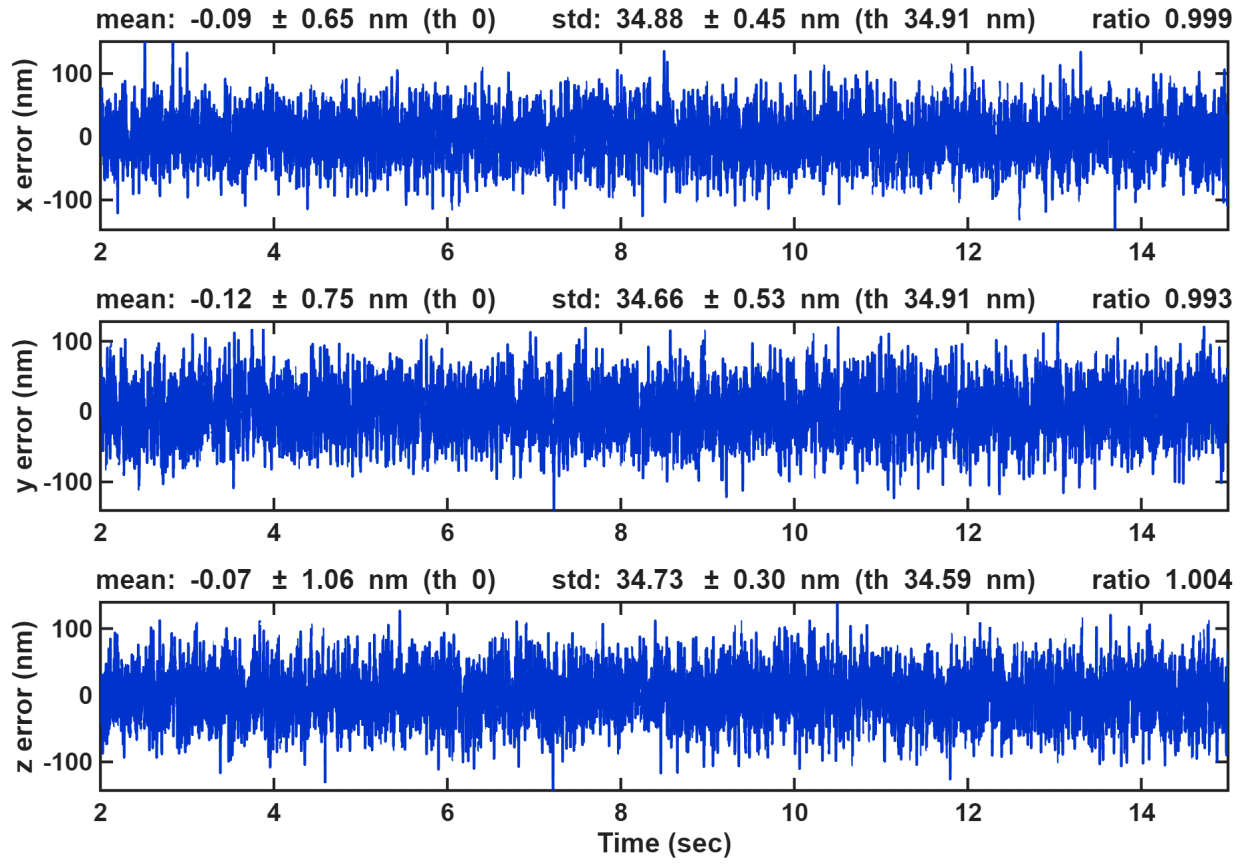


Figure 1: Tracking error per axis ($h = 50 \mu\text{m}$).

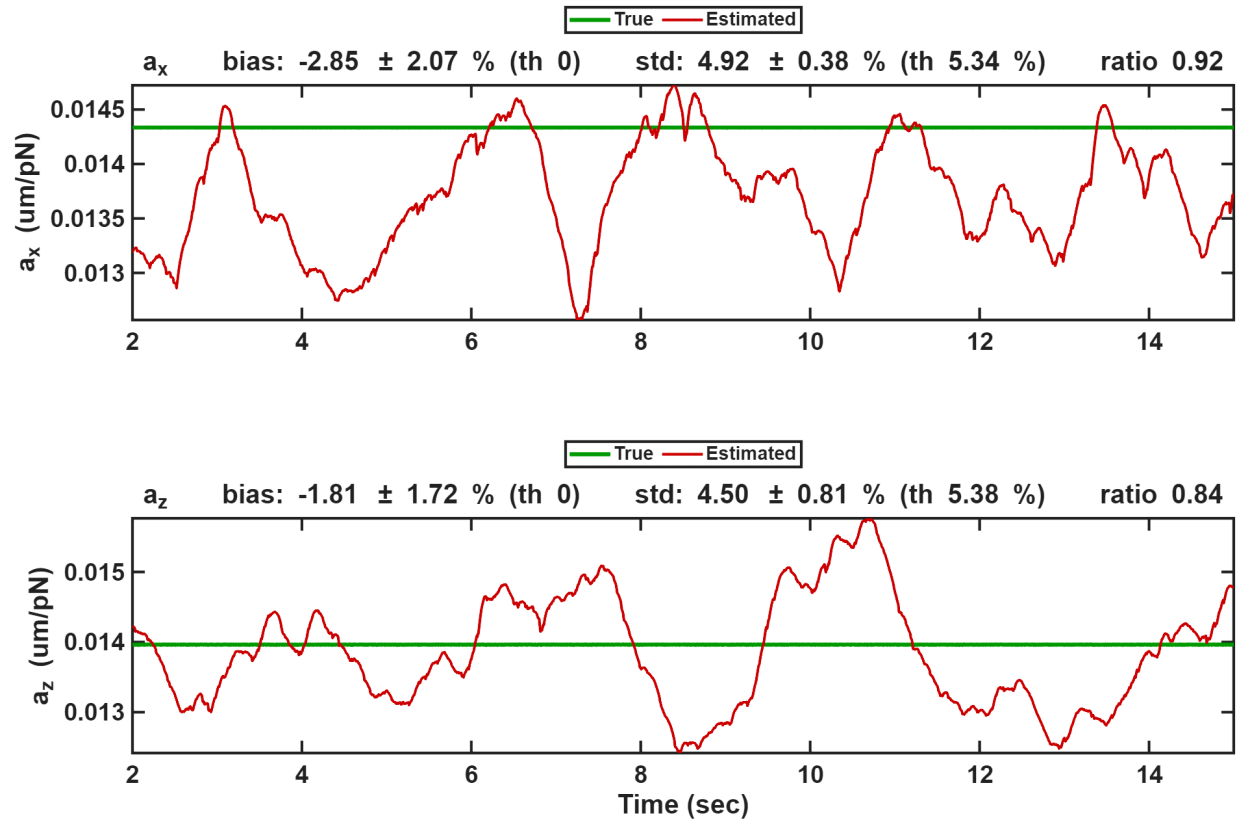


Figure 2: Motion-gain estimation ($h = 50 \mu\text{m}$).